Reg. No.															

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Fourth and Fifth Semester

21MAB302T - DISCRETE MATHEMATICS

(For the candidates admitted from the academic year2021-2022 to 2023 - 2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
1.	If $f : A \to B$ and $g : B \to C$ are bijections then $g \circ f$ is A) a bijection B) not a bijection C) only injective D) only surjective	1	2	1	2
2.	If $\|A\|=6$, then the cardinality of power set of A is A) 64 B) 8 C) 32 D) 16	1	2	1	2
3.	If M_R represents the matrix relation on a set A such that $M_R = MRT$ then the relation is A) symmetric B) reflexive C) transitive D) irreflexive	1	1	1	2
4.	Which of the following forms a partition of the set A={1,2,3} ? A) { {1,2},{1,3} } B) { {1,2,3},{3} } C) { {1},{2,3} } D) { {1,2},{2,3} }	1	2	1	2
5.	The minimum number of persons in a group to be sure that atleast two of them are born in the same month is A) 12 B) 13 C) 24 D) 25	1	2	2	2
6.	Number of ways in which 7 persons can be seated round a table is A) 120 B) 720 C) 540 D) 480	1	2	2	2
7.	Every positive integer n > 1 can be written uniquely as a product of A) Positive integer B) Perfect square C) Perfect cube D) Primes	1	1	2	2
8.	If a prime number p divides a^2, then A) p divides a B) a divides p C) p does not divide a D) a does not divide p	1	1	2	2
9.	The converse of $q \to p$ is A) $p \to \neg q$ B) $\neg p \to \neg q$ C) $p \to q$ D) $\neg p \to q$	1	1	3	2

10. The dual of $Q \vee F \equiv Q$ is — 1 1 3 2

A) $Q \vee T \equiv Q$ B) $Q \vee F \equiv T$
C) Q D) $Q \wedge T \equiv Q$

11. The set of premises is said to be _____ if their conjunction implies a contradiction. — 1 1 3 2

A) Tautology B) Inconsistent
C) Consistent D) Universe of discourse

12. $p \rightarrow (q \rightarrow p) \equiv$ — 1 2 3 2

A) T B) F
C) both T and F D) neither T nor F

13. In a Multiplicative group $G = \{1, -1, i, -i\}$, order of the element i is — 1 2 4 2

A) 3 B) 4
C) 2 D) 1

14. Which of the following statement is true? — 1 2 4 2

A) Every group is abelian B) Every abelian group is cyclic
C) Every cyclic group is abelian D) Every group is cyclic

15. Every group of prime order is — 1 1 4 2

A) not cyclic B) cyclic
C) left coset D) right coset

16. The Hamming distance H(x, y) between the codes x = 11010 and y = 10101 is — 1 2 4 1

A) 1 B) 2
C) 3 D) 4

17. A simple graph in which there is exactly one edge between each pair of distinct vertices is called a — 1 1 5 2

A) Regular graph B) Bipartite graph
C) Complete graph D) Pseudo graph

18. A tree with 8 edges has — 1 2 5 2

A) 6 Vertices B) 7 Vertices
C) 8 Vertices D) 9 Vertices

19. The maximum number of edges in a complete graph with 5 vertices (K_5) is — 1 2 5 2

A) 10 B) 12
C) 16 D) 20

20. The number vertices of odd degree in undirected graph is — 1 1 5 2

A) Zero B) One
C) Even D) Odd

PART - B (5 x 8 = 40 Marks) — Marks BL CO PO
Answer **ALL** Questions

21 a. Using Warshalls algorithm, find the transitive closure of the relation defined on the set A = {1,2,3,4} , — 8 4 1 1

R = {(1,2), (2,1), (2,3), (3,4), (4,1)} .

(OR)

b. If $f : A \to B$ and $g : B \to C$ are invertible functions, then prove $g \circ f : A \to C$ is also invertible and also prove that $(g \circ f)^{-1} = (f^{-1} \circ g^{-1})$. | 8 | 4 | 1 | 2

22 a. Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2,3,5 and 7. | 8 | 3 | 2 | 2

(OR)

b. Use the Euclidean algorithm to find GCD *(3587,1819)* and also express the gcd as a linear combination of the given two numbers . | 8 | 3 | 2 | 1

23 a. Using indirect method of proof, derive $p \to \neg s$ from the premises $p \to q \vee r$, $q \to \neg p$, $s \to \neg r$, p. | 8 | 3 | 3 | 1

(OR)

b. i) Construct the truth table for $(P \to (Q \to R)) \to ((P \to Q) \to (P \to R))$. (4 Marks) | 8 | 3 | 3 | 2

ii) Prove the equivalence without using truth tables $p \to (q \to p) \equiv \neg p \to (p \to q)$. (4 Marks)

24 a. Show that the set of all positive rational numbers forms an abelian group under the binary operation * defined by | 8 | 3 | 4 | 2

$a * b = (ab)/2$, for $a, b \in Q^+$.

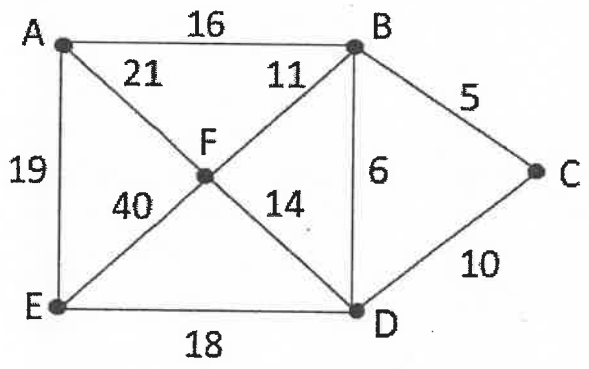
(OR)

b. Prove that every subgroup of a cyclic group is again cyclic. | 8 | 3 | 4 | 2

25 a. Prove that the maximum number of edges in a bipartite graph with n vertices is atmost $(n/2)^2$. | 8 | 3 | 5 | 2

(OR)

b. Find the minimum spanning tree of the following weighted graph using Kruskal's algorithm . Also find the minimum total weight of the spanning tree. | 8 | 3 | 5 | 1



PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Question

Marks BL CO PO

26. a) Define Partially ordered set. (3 Marks) | 15 | 4 | 1 | 2

b) Prove that the relation $\subseteq$ of set inclusion is a partial ordering relation on the power set $P(S)$, where $S = \{ a,b,c \}$. Also draw the Hasse diagram representing the above poset. (12 Marks)

27. a) Find the minimum distance between the codes x = 10110, y = 11110 , z = 10011. 15 4 4 1
(3 Marks)

b) Find the code words generated by the parity check matrix.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

when the encoding function is $e: B^3 \rightarrow B^6$. (12 Marks)

******